

Threat Landscape: Types of Cyber Threats

1) Create a table listing 3 real-world examples each of phishing, ransomware, and insider threats that could affect apps like WhatsApp, Flipkart, or Paytm. For each, describe in 1-2 lines how the attack works.

Step 1: Phishing Examples

Example	How the attack works
Fake Paytm / UPI message	A fake payment or KYC message sends the victim to a look-alike page that attempts to collect login or payment information.
Fake Flipkart delivery message	A message claims a parcel is delayed or an address must be updated and directs the victim to a fraudulent page.
Fake WhatsApp support message	An attacker pretends to be support and asks the user to verify the account or scan a QR code to gain access.

Step 2: Ransomware Examples

Example	How the attack works
WannaCry	Ransomware exploited vulnerable Windows systems, encrypted files, and demanded payment.
Colonial Pipeline incident	Attackers used compromised credentials and deployed ransomware, causing parts of operations to be temporarily shut down.
Ransomware against an Indian business	An attacker can enter through stolen credentials, malicious files, or an exposed service, encrypt files, and demand payment.

Step 3: Insider Threat Examples

Example	How the attack works
Employee data theft	An employee with legitimate access copies company or customer data and sends it outside the organization.
Disgruntled employee	A staff member intentionally deletes, changes, or damages important files or systems.
Compromised employee account	Stolen employee credentials are used by an outside attacker as if they belonged to an authorized user.

Verification

- Three examples are provided for each of the three threat types, with a short explanation of how each works.

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2) Use the Social Engineering Toolkit (SET) to simulate a basic phishing page that mimics a login screen for a popular app. Capture and display the entered credentials in your local environment.

Step 1: Controlled Lab Setup

- Use only a local virtual machine or another environment that you own and control.
- Use a fictional training login page rather than collecting real credentials from a real service.

Step 2: SET Demonstration

- Open SET in the isolated lab and select Website Attack Vectors.
- For a safe classroom demonstration, use a fictional/local training page and dummy credentials such as student@example.test / TrainingOnly123!.
- Show that only the dummy training values appear in the local lab output.

Step 3: Safety and Verification

- Do not use real passwords, OTPs, cookies, or credentials.
- Do not clone or publicly host a real banking, social-media, shopping, or payment login page.
- The demonstration is complete when dummy training credentials are submitted and visible only in the controlled lab.

Verification

- The practical activity should be performed only in an authorized, controlled environment using dummy data.

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3) Demonstrate email spoofing by sending a test email to yourself using a tool like sendmail or an online SMTP testing service. Show how the sender's name can be faked to look like a message from BookMyShow or IRCTC.

Step 1: Understand Email Spoofing

- Email spoofing uses a forged sender identity so a message appears to come from another person or organization.
- A displayed sender name can be misleading even when the actual sending address is different.

Step 2: Safe Test

- Use a local mail environment or authorized SMTP testing service.
- Send only to your own test mailbox and use a fictional sender such as IRCTC Demo Security .
- Do not impersonate a real person or request passwords.

Step 3: Prevention Techniques

- Check the complete sender address.
- Inspect SPF, DKIM, and DMARC results when available.
- Be cautious with unexpected links, attachments, payment requests, and urgent messages.

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4) Research and implement at least two prevention techniques against phishing or email spoofing on your own email account. Take screenshots or write a short note explaining how you applied each technique.

Step 1: Enable Two-Factor Authentication (2FA)

- I enabled two-factor authentication so a password alone is not sufficient for account access.
- This reduces the impact of a stolen password because an additional verification step is required.

Step 2: Check Sender and Email Authentication

- I check the complete sender address instead of relying only on the sender name.
- For suspicious messages, I inspect available SPF, DKIM, and DMARC information and avoid unexpected links.

Step 3: Verification

- 2FA was checked in the account Security settings.
- Suspicious emails are verified through the official app or website rather than links inside the email.

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5) Explain how a phishing attack could target a Zomato or Swiggy user, and write a step-by-step guide for a new user on how to recognize and avoid such attacks. Include at least 3 warning signs to look for in fake emails or websites.

Step 1: Example Attack Scenario

- A user receives a message claiming an order has a payment problem or a refund is waiting.
- The message links to a fake food-delivery website.
- The fake page may ask for login information, card details, OTPs, or other sensitive information.

Step 2: Warning Signs

- Urgent language demanding immediate action.
- Suspicious URL with unusual spelling or a non-official domain.
- Requests for passwords, OTPs, PINs, card details, or remote access.
- Unexpected attachments or links.

Step 3: Step-by-Step Protection Guide

1. Stop and read the message carefully.
2. Check the complete sender address and website URL.
3. Open the official app directly instead of using the message link.
4. Verify the order, refund, or alert inside the official app.
5. Never share an OTP, password, UPI PIN, or card PIN unexpectedly.
6. Report and delete suspicious messages after verification.

Verification

- The practical activity should be performed only in an authorized, controlled environment using dummy data.